

A Projection Classification for Monod's Operator Alpha-Ratio and Alpha-Reiter Problems

Run `fa_banach.001`, lane 7

June 20, 2026

Source Problems

The source paper is N. Monod, *Fixed points in convex cones*, arXiv:1701.05537. Near the end of the paper, after discussing ℓ^p -variants of the f -ratio and f -Reiter properties, the author introduces the following Hilbert-space analogues. A unitary representation π of a group G on a Hilbert space V has the operator α -ratio property, for a continuous operator $\alpha \neq 0$, if for every finite $S \subseteq G$ and every $\varepsilon > 0$ there is $v \in V$ such that

$$\|\alpha\pi(s)v\| < (1 + \varepsilon)\|\alpha v\| \quad (s \in S).$$

It has the stronger operator α -Reiter property if one can choose v such that

$$\|\alpha(\pi(s)v - v)\| < \varepsilon\|\alpha v\| \quad (s \in S).$$

Problems 47 and 48 ask to clarify these properties, especially for nonzero projections α .

straightforward verification where f , u and ϵ are to be replaced with appropriate powers. The same is true for the f -Reiter property using basic L^p inequalities.

The particular case of $p = 2$ suggests the following:

We say that a unitary representation of a group G on a Hilbert space V has the **operator α -ratio property** for a continuous operator $\alpha \neq 0$ of V if for every finite set $S \subseteq G$ and every $\epsilon > 0$, there is $v \in V$ with $\|\alpha(sv)\| < (1 + \epsilon)\|\alpha(v)\|$ for all $s \in S$. Likewise, the representation has the stronger **operator α -Reiter property** if we can choose v with $\|\alpha(sv - v)\| < \epsilon\|\alpha(v)\|$ for all $s \in S$.

Problem 47. Clarify under which circumstances the operator α -ratio and α -Reiter properties hold.

Returning closer to supramenability:

Problem 48. Characterise the unitary representations that satisfy the operator α -ratio or α -Reiter property for all non-zero projections α in V .

Proof Intuition

The ratio property is much weaker than it first appears. It only compares the α -size of finitely many unitary translates with the original α -size. A vector almost attaining the operator norm of α already works, so the ratio property is automatic for every bounded nonzero α .

The projection-Reiter side has an equally sharp obstruction. If the representation is nontrivial, then some unitary operator $U = \pi(g)$ is not the identity. The spectral theorem gives a nonzero spectral subspace on which U stays a definite distance away from 1. Projecting onto that spectral subspace turns the Reiter inequality into a uniform lower bound, so Reiter fails for that projection. Conversely, a trivial representation has zero movement and hence satisfies every Reiter condition.

Theorem 1. *Let $\pi : G \rightarrow \mathcal{U}(V)$ be a unitary representation on a Hilbert space.*

1. *For every nonzero $\alpha \in B(V)$, the representation has the operator α -ratio property.*
2. *For every nonzero $\alpha \in B(V)$, the representation has the operator α -Reiter property if and only if, for every finite $S \subseteq G$,*

$$\inf_{\alpha v \neq 0} \frac{\sum_{s \in S} \|\alpha(\pi(s)v - v)\|^2}{\|\alpha v\|^2} = 0.$$

Equivalently, for no finite $S \subseteq G$ does there exist $c > 0$ such that

$$\sum_{s \in S} \|\alpha(\pi(s)v - v)\|^2 \geq c \|\alpha v\|^2 \quad (v \in V).$$

3. *The representation has the operator P -Reiter property for every nonzero orthogonal projection P on V if and only if π is trivial.*

Proof. For the ratio property, fix finite $S \subseteq G$ and $\varepsilon > 0$. Since $\alpha \neq 0$, choose a unit vector v with

$$\|\alpha v\| > \frac{\|\alpha\|}{1 + \varepsilon}.$$

For each $s \in S$, unitarity gives

$$\|\alpha\pi(s)v\| \leq \|\alpha\| < (1 + \varepsilon)\|\alpha v\|.$$

This proves (1).

For (2), the displayed infimum is zero exactly when, for every $\delta > 0$, there exists v with $\alpha v \neq 0$ and

$$\sum_{s \in S} \|\alpha(\pi(s)v - v)\|^2 < \delta^2 \|\alpha v\|^2.$$

Taking $\delta = \varepsilon$ gives the defining Reiter inequalities for all $s \in S$. Conversely, the defining inequalities imply the displayed sum estimate with $\delta^2 = |S|\varepsilon^2$. The second formulation is simply the negation of the existence of a positive lower bound for this Rayleigh quotient.

For (3), first suppose that π is trivial. Then $\alpha(\pi(s)v - v) = 0$ for every operator α , every s , and every v , so every projection-Reiter condition holds.

Conversely, suppose π is not trivial. Choose $g \in G$ such that $U = \pi(g) \neq I$. By the spectral theorem for the unitary U , there is $\delta > 0$ such that the spectral projection

$$P = 1_{\{z \in \mathbb{T} : |z-1| \geq \delta\}}(U)$$

is nonzero. This projection commutes with U , and on PV the operator $U - I$ is bounded below by δ :

$$\|(U - I)w\| \geq \delta \|w\| \quad (w \in PV).$$

Therefore, for every $v \in V$,

$$\|P(\pi(g)v - v)\| = \|(U - I)Pv\| \geq \delta\|Pv\|.$$

Taking the finite set $S = \{g\}$ and any $0 < \varepsilon < \delta$ shows that the operator P -Reiter property fails. Hence a representation satisfying P -Reiter for every nonzero orthogonal projection must be trivial. $\square$

Corollary 1 (Problem 48). *For unitary representations on Hilbert space:*

1. *every representation satisfies the operator P -ratio property for all nonzero projections P ;*
2. *the representations satisfying the operator P -Reiter property for all nonzero projections P are exactly the trivial representations.*

The same conclusion holds if “projection” is read as arbitrary bounded idempotent rather than orthogonal projection.

Proof. The ratio statement is Theorem 1(1). For Reiter, orthogonal projections are among arbitrary bounded idempotents, so the spectral-projection obstruction applies under either convention. A trivial representation has zero displacement and therefore satisfies the Reiter inequalities for every operator, not only projections. $\square$

Corollary 2 (Bounded-below operators). *If α is bounded below, then the operator α -Reiter property is equivalent to the ordinary existence of almost invariant vectors for π .*

Proof. Choose constants $0 < m \leq M < \infty$ with

$$m\|w\| \leq \|\alpha w\| \leq M\|w\| \quad (w \in V).$$

If π has almost invariant vectors, choose $v \neq 0$ with

$$\|\pi(s)v - v\| < \varepsilon m M^{-1} \|v\| \quad (s \in S).$$

Then

$$\|\alpha(\pi(s)v - v)\| \leq M\|\pi(s)v - v\| < \varepsilon m \|v\| \leq \varepsilon \|\alpha v\|.$$

Conversely, if α -Reiter holds, choose v with

$$\|\alpha(\pi(s)v - v)\| < \varepsilon m M^{-1} \|\alpha v\| \quad (s \in S).$$

Then

$$m\|\pi(s)v - v\| \leq \|\alpha(\pi(s)v - v)\| < \varepsilon m M^{-1} \|\alpha v\| \leq \varepsilon m \|v\|,$$

which gives ordinary almost invariance. $\square$

Corollary 3 (Finite-rank projections). *Let P be an orthogonal projection with finite-dimensional range W . Then π has the operator P -Reiter property if and only if, for every finite $S \subseteq G$, there exists $v \in V$ such that*

$$Pv \neq 0 \quad \text{and} \quad P(\pi(s)v - v) = 0 \quad (s \in S).$$

In particular, for a rank-one projection P_ξ onto the line spanned by $0 \neq \xi \in V$, this is equivalent to

$$\xi \notin \text{span}\{\pi(s)^{-1}\xi - \xi : s \in S\} \quad \text{for every finite } S \subseteq G.$$

Proof. For fixed S , define

$$L : V \rightarrow W \oplus W^S, \quad L(v) = (Pv, (P(\pi(s)v - v))_{s \in S}).$$

Since the target is finite-dimensional, $L(V)$ is closed. If the exact condition holds, then the P -Reiter inequalities hold with zero left hand side. Conversely, if P -Reiter holds, choose v_n with $\|Pv_n\| = 1$ and

$$\sum_{s \in S} \|P(\pi(s)v_n - v_n)\|^2 \rightarrow 0.$$

After passing to a subsequence, Pv_n converges to some unit vector $w \in W$. Hence $(w, 0)$ lies in the closure of $L(V)$ and therefore in $L(V)$, giving the exact condition.

For the rank-one statement, the equations become

$$\langle v, \xi \rangle \neq 0, \quad \langle v, \pi(s)^{-1}\xi - \xi \rangle = 0 \quad (s \in S),$$

which is possible exactly when ξ does not belong to the span displayed above. □

Scope and Novelty Check

The theorem gives a complete answer to Problem 48 under both common projection conventions. It also gives an exact spectral-gap criterion for the general operator α -Reiter side of Problem 47. The broader group-theoretic questions in the same source paper about products, supramenability, and finitely generated exponential-growth examples with the fixed-point property for cones are separate and are not addressed here.

The local run indexes were searched for arXiv:1701.05537 and phrases including `operator alpha-ratio`, `operator alpha-Reiter`, `projection alpha classification`, and `fixed points in convex cones`. Web searches for exact phrases such as `"operator alpha-ratio"`, `"operator alpha-Reiter"`, `"operator alpha-Reiter"`, `"projection"`, and `"Fixed points in convex cones"` `"Problem 48"` found the source paper but no later explicit answer to Problems 47–48.

References

- [1] N. Monod, *Fixed points in convex cones*, arXiv:1701.05537, 2017.